SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS **COURSE CODE: MLA0102**

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 45 %

Total Marks: 50 **Time: 60 Mins**

No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I D. Sri Anjaneya / 192425219 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	

Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty Total Marks Awarded

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	**Healthcare**	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	**Finance**	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	**Retail**	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	**Autonomous Vehicles**	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	**Robotics**	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.
6	**Manufacturing**	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

Structure of the Report:

S.No	Report Component	Student Deliverable / Guiding Question
1	**Title of AI Application**	Provide the name of the latest AI application selected and the corresponding AI domain.
2	**Domain Selection**	Identify the application domain (Healthcare / Finance / Retail / Autonomous Vehicles / Robotics / Manufacturing).
3	**Introduction and Background**	Describe the AI application and provide an overview of its purpose and importance.
4	**Problem Identification**	Explain the real-world problem or challenge addressed by this AI application.
5	**Need for AI Solution**	Analyze why AI is required and how it improves traditional approaches.
6	**AI Technologies Used**	Identify the AI techniques involved (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Robotics, etc.).
7	**Working Mechanism**	Explain how the AI system works, including input data, processing methods, decision-making, and output generation.
8	**Data Sources and Processing**	Describe the type of data used and how the data is processed for intelligent decision-making.
9	**Real-World Implementation**	Identify organizations, industries, or real-world scenarios where this AI application is currently implemented.
10	**Impact and Benefits**	Evaluate the improvements achieved through the AI application in terms of efficiency, accuracy, cost, user experience, or automation.
11	**Challenges and Limitations**	Discuss technical, ethical, privacy, security, and deployment challenges associated with the application.
12	**Future Scope and Emerging Trends**	Explain possible enhancements, future developments, and research opportunities in this AI application.
13	**Conclusion**	Summarize the significance of the AI application and its contribution to solving real-world problems.
14	**References**	Provide research papers, technical articles, industry reports, and reliable sources referred for the study.

Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI application.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanation with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplete.
3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanation with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriate citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.
5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation with appropriate examples.	Good implementat ion analysis.	Basic implementation discussion.	Minimal or no real-world analysis.
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluation with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical evaluation.	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.
8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.

AI-Powered Autonomous Driving: A Study of Waymo's Self-Driving Technology

AI Domain: Autonomous Vehicles

1. Domain: Autonomous Vehicles

Autonomous Vehicles is a field of Artificial Intelligence that focuses on developing vehicles capable of understanding their surroundings and performing driving tasks with minimal or no human intervention.

This report focuses on **Waymo's Self-Driving Technology**, which uses AI algorithms, sensors, and intelligent decision-making systems to navigate roads, detect obstacles, recognize pedestrians, and respond to changing traffic conditions.

The domain was selected because autonomous driving is an emerging real-world application of AI that has the potential to improve transportation safety, accessibility, and efficiency while introducing important technical and ethical challenges.

2. Introduction and Background

Autonomous vehicles are vehicles that use Artificial Intelligence (AI), sensors, and intelligent computing systems to perform driving tasks with minimal or no human intervention. Unlike traditional vehicles, which depend on human drivers to observe road conditions and make decisions, autonomous vehicles use AI to analyze their surroundings and respond to changing traffic situations.

The development of autonomous vehicles has become an important area of AI research because driving involves complex tasks such as identifying pedestrians, recognizing traffic signals, detecting obstacles, and predicting the movements of other vehicles.

Waymo's Self-Driving Technology is a real-world application of AI in autonomous transportation. It combines sensor data and AI-based software to understand road conditions, make driving decisions, and navigate supported routes.

This technology has the potential to improve transportation safety, provide convenient mobility services, and increase accessibility for people who face difficulties driving independently. However,

challenges such as unexpected road situations, adverse weather, system reliability, and safety validation remain important considerations.

3. Problem Identification

Traditional vehicles depend on human drivers to observe road conditions, recognize traffic signals, identify obstacles, and make timely driving decisions. However, human errors caused by distraction, fatigue, and delayed reactions can contribute to road accidents. Driving in heavy traffic and responding to unexpected situations can also be challenging.

The major problems addressed by AI-powered autonomous vehicles are:

1. **Human Error:** Distraction, tiredness, and delayed reactions may affect driving safety.
2. **Complex Traffic Conditions:** Drivers must continuously monitor vehicles, pedestrians, cyclists, and traffic signals.
3. **Unexpected Obstacles:** Sudden braking, road obstructions, and unpredictable pedestrian movements require quick responses.
4. **Transportation Accessibility:** Some people face difficulties driving independently and may need alternative mobility services.
5. **Real-Time Decision-Making:** Vehicles must quickly analyze changing road conditions and select appropriate actions.

The main problem is to develop a reliable system that can understand the road environment, predict potential hazards, and make safe driving decisions in real time.

Waymo's self-driving technology addresses this problem by using AI algorithms and sensor systems to perceive the surroundings, predict the movements of other road users, and plan appropriate driving actions. However, autonomous driving systems can still make mistakes, so extensive testing and safety measures are necessary.

4. Need for AI Solution

Artificial Intelligence is needed in autonomous vehicles because driving requires continuous observation, quick decision-making, and the ability to respond to changing road conditions. Traditional vehicles mainly depend on human drivers, while autonomous vehicles use AI algorithms and sensors to perform these tasks automatically.

The major reasons for using AI in autonomous vehicles are:

1. **Real-Time Object Detection:** AI helps identify vehicles, pedestrians, traffic signals, and obstacles on the road.
2. **Faster Decision-Making:** AI systems process sensor information and select suitable driving actions according to the surrounding conditions.
3. **Prediction of Movement:** AI helps estimate the possible movements of nearby vehicles and pedestrians, allowing the system to prepare for potential hazards.
4. **Improved Environmental Understanding:** Combining data from different sensors helps the vehicle understand road conditions more effectively.
5. **Automated Driving:** AI-based planning and control systems help manage steering, acceleration, and braking with minimal human intervention.
6. **Potential Safety Improvement:** Continuous environmental monitoring can help reduce certain risks associated with driver distraction and delayed reactions.

Therefore, AI provides the perception, prediction, and decision-making capabilities required for autonomous driving. In Waymo's self-driving technology, these capabilities support navigation and responses to changing traffic conditions. However, AI cannot eliminate all driving risks, and reliable operation requires extensive testing, safety validation, and appropriate monitoring.

5. AI Technologies Used

Waymo's self-driving technology uses a combination of Artificial Intelligence, sensors, and intelligent computing systems to understand road conditions and make driving decisions. The following AI technologies and computational methods are important in autonomous driving.

1. Computer Vision

Computer vision enables the vehicle to analyze camera images and identify road users, traffic signals, road signs, and obstacles. It helps the system understand the visual environment.

2. Deep Learning

Deep learning uses neural networks trained on large amounts of data to recognize complex patterns in driving scenes. It supports tasks such as object detection, scene understanding, and movement prediction.

3. Sensor Fusion

Sensor fusion combines information from cameras, LiDAR, and radar. Cameras provide visual information, LiDAR measures distances and surrounding geometry, and radar helps estimate the distance and relative movement of objects. Combining these sources provides a more comprehensive view of the road.

4. Object Detection and Tracking

Object detection identifies vehicles, pedestrians, cyclists, and other obstacles. Object tracking monitors their positions and movements over time, helping the vehicle understand how the surrounding environment is changing.

5. Motion Prediction

Motion prediction estimates the possible future movements of nearby road users. For example, the system may predict whether a pedestrian is likely to cross the road or another vehicle may enter its path.

6. Path Planning and Decision-Making

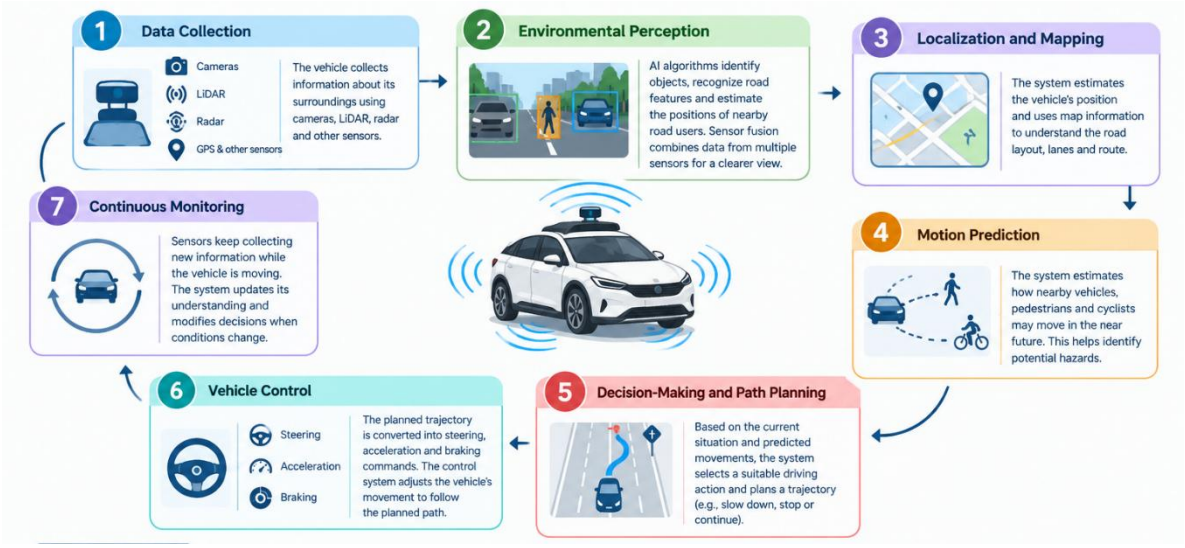
Path planning determines a suitable trajectory based on road conditions, traffic rules, obstacles, and predicted movements. Decision-making selects appropriate actions, such as slowing down, stopping, or proceeding safely.

7. Vehicle Control Systems

Vehicle control systems convert the planned trajectory into steering, acceleration, and braking commands. Feedback from vehicle sensors helps the system adjust its movement continuously.

These technologies work together to support autonomous driving. Computer vision and sensor fusion help the vehicle understand its surroundings, prediction estimates possible future situations, and planning and control determine and execute suitable driving actions. The exact internal implementation of Waymo's proprietary system may differ from this general description.

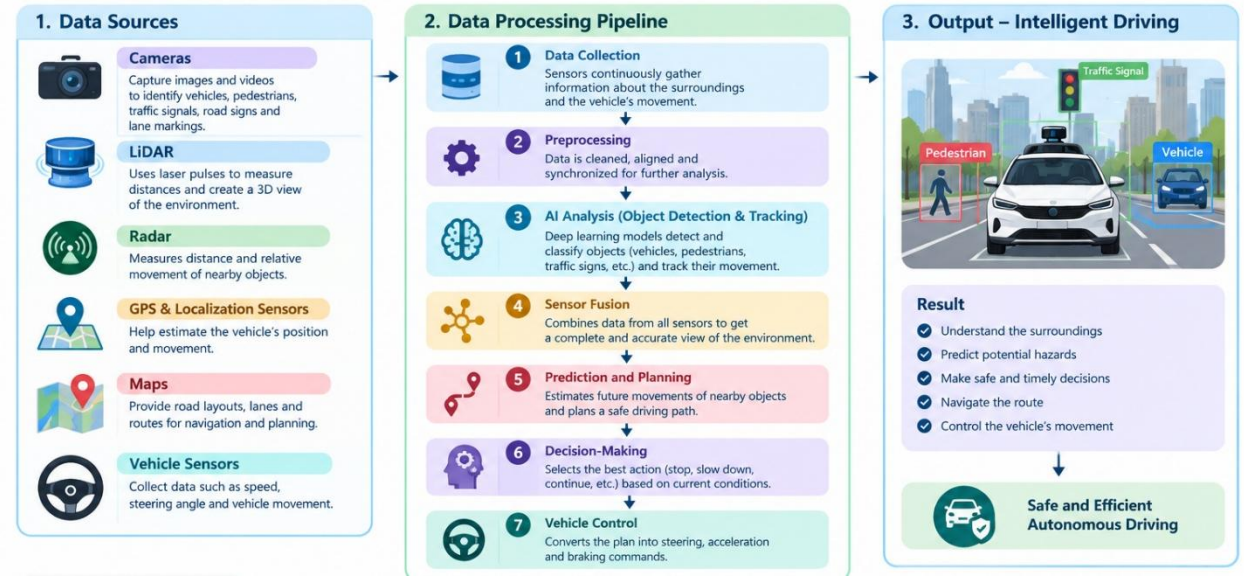
6. Working Mechanism



7. Data Sources and Processing

1. Data Sources: Cameras, LiDAR, radar, GPS, maps, and vehicle sensors collect information about the surroundings.
2. Data Collection: The system gathers sensor measurements continuously.
3. Preprocessing: The collected data is cleaned, aligned, and synchronized.
4. AI Analysis: Deep-learning models detect and track vehicles, pedestrians, and traffic signals.
5. Sensor Fusion: Information from different sensors is combined to improve environmental understanding.
6. Prediction and Planning: The system predicts possible movements and plans a suitable driving path.

7. Decision-Making and Vehicle Control: It selects actions such as stopping or slowing down and converts the driving plan into steering, acceleration, and braking commands.



8. Real-World Implementation

Case Study: Waymo's Self-Driving Technology

Waymo is a company that develops autonomous driving technology and provides self-driving ride-hailing services in selected areas. Its autonomous driving system, known as the Waymo Driver, uses Artificial Intelligence, sensors, and specialized software to navigate roads and transport passengers without a human driver controlling the vehicle.

Working in the Real World

1. **Ride Booking:** Passengers request a ride through the Waymo One mobile application in supported service areas.
2. **Vehicle Arrival:** An autonomous vehicle arrives at the passenger's pickup location.
3. **Environmental Perception:** Cameras, LiDAR, radar, and AI software help the vehicle detect pedestrians, vehicles, traffic signals, and obstacles.

4. **Navigation and Decision-Making:** The system follows a planned route and responds to changing traffic conditions by selecting appropriate driving actions.
5. **Passenger Drop-Off:** The vehicle transports the passenger to the destination while continuously monitoring its surroundings.

Real-World Significance

Waymo demonstrates how autonomous driving technology can be used in commercial transportation rather than remaining limited to research and laboratory testing. Its services provide a practical example of AI being applied to navigation, environmental perception, and automated vehicle control.

However, autonomous driving still has limitations. Unexpected road situations, adverse weather, sensor errors, and complex traffic conditions can affect system performance. Therefore, safety testing, operational restrictions, and regulatory compliance are essential for responsible deployment.

This case study shows that autonomous vehicles have practical applications in transportation, while also highlighting the need for continued research and safety evaluation.

9. Impact and Benefits

AI-powered autonomous vehicles have the potential to improve transportation by using intelligent systems to understand road conditions, make driving decisions, and control vehicle movement. The major impacts and benefits are explained below.

1. Improved Road Safety

AI systems continuously monitor the surrounding environment and can identify potential hazards. They may help reduce certain accidents caused by human distraction, fatigue, or delayed reactions. However, autonomous vehicles cannot eliminate all road accidents.

2. Increased Accessibility

Autonomous ride-hailing services may provide additional transportation options for elderly people and individuals with disabilities who face difficulties driving independently, depending on service availability and accessibility features.

3. Reduced Human Effort

Passengers can travel without personally controlling the vehicle. This may reduce the effort required for driving and allow passengers to use their travel time for other activities.

4. Improved Traffic Efficiency

AI systems can analyze traffic conditions and select suitable routes. Efficient route planning and smoother driving may help reduce unnecessary delays in some situations.

5. Continuous Environmental Monitoring

Unlike human drivers, AI systems do not experience human fatigue. They can repeatedly process sensor information to monitor vehicles, pedestrians, traffic signals, and obstacles. However, technical failures and incorrect predictions remain possible.

6. Technological Development

Autonomous vehicles encourage advancements in computer vision, deep learning, robotics, sensor technology, and intelligent transportation systems.

7. Potential Environmental Benefits

Autonomous driving systems may support smoother driving and efficient route selection, potentially reducing unnecessary energy consumption. The actual environmental impact depends on vehicle type, traffic conditions, electricity sources, and travel demand.

In conclusion, AI-powered autonomous vehicles offer potential benefits in road safety, accessibility, convenience, and transportation efficiency. However, these benefits depend on reliable system performance, appropriate safety measures, and responsible implementation.

10.Challenges and Limitations

1. **Weather Conditions:** Heavy rain, fog, and poor visibility can affect sensors and reduce the accuracy of environmental perception.
2. **Unexpected Situations:** Construction zones, unusual obstacles, and unpredictable pedestrian movements can make driving decisions difficult.

3. **High Costs:** Advanced sensors, computing hardware, software development, and testing increase the cost of autonomous vehicles.
4. **Safety and Cybersecurity:** AI errors or cyberattacks may affect vehicle operation, making rigorous testing and security measures essential.
5. **Privacy and Legal Issues:** Sensors may collect personal information, while responsibility for accidents involving autonomous vehicles can raise legal and ethical questions.

11.Future Scope and Emerging Trends

1. **Advanced AI Models:** Improved deep learning models may help vehicles understand complex road situations and make better driving decisions.
2. **Improved Sensor Technology:** More accurate cameras, LiDAR, and radar systems may help vehicles detect obstacles in different environments.
3. **Vehicle-to-Everything (V2X) Communication:** Vehicles may communicate with traffic signals, other vehicles, and road infrastructure to improve traffic coordination.
4. **Autonomous Delivery Services:** Self-driving vehicles and delivery robots may transport food, parcels, and essential goods with reduced human involvement.
5. **Smart City Integration:** Autonomous vehicles may work with intelligent traffic management systems to support smoother transportation.
6. **Better Safety and Testing:** Advanced simulation, real-world testing, and improved safety systems may help identify risks before wider deployment.

12.Conclusion

AI-powered autonomous vehicles represent an important development in modern transportation. This report studied Waymo’s self-driving technology and explained how artificial intelligence, computer vision, deep learning, sensor fusion, and intelligent decision-making help vehicles understand their surroundings and navigate roads.

Autonomous driving has the potential to improve road safety, support people with limited mobility, and make transportation more convenient. However, challenges such as unpredictable road conditions, high development costs, cybersecurity risks, privacy concerns, and legal responsibilities must be addressed.

In conclusion, autonomous vehicles are a promising application of artificial intelligence. With continuous research, extensive safety testing, improved technology, and appropriate regulations, systems such as Waymo's may contribute to safer and more efficient transportation in the future.

13.References

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